

Learning Plan - Aggregator

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1. What needs to be learned
 - a. What is an HTTP server?
 - i. How does hosting a web server work?
 - b. What is an API?
 - i. How do API limits work?
 - ii. What is an API key and how does it work?
 - iii. How do you deal with API request failures?
 - c. How to use AWS/Render?
 - d. How does GitHub actions work?
 - e. What is CI/CD?
2. Who learns each thing, from where, and by what date?
 - a. Everyone learns what an API and HTTP server are
 - b. The person whose job it is to create tests needs to learn GitHub actions and what CI/CD is.
 - c. The people who will set up the web server need to learn how to use AWS and what a web server is and how to set one up. Learning Sources
 - d. What is an API? [Introduction to API \(Application Programming Interface\) - GeeksforGeeks](#)

i. How do API limits work?

Approach	Description	Key characteristics	Use cases
Key-level rate limiting	Controls API traffic based on the unique API key of the user. Can be global (for all APIs) or per API.	– Rate limits based on individual API keys – Can set global or per-API limits – Ensures fairness per user	– Managing traffic per user or API key – Freemium model APIs – Balancing API usage across multiple services
API-level rate limiting	Sets a rate limit based on the total traffic entering an API from all sources.	– Limits traffic across all users accessing a specific API – Focused on overall API health – Simple to implement	– Preventing overload on a specific API – Ensuring API scalability – Handling unexpected traffic spikes
User-based rate limiting	Limits requests based on specific users, ensuring individual user traffic is controlled.	– Different thresholds for different users – Can be tied to account tiers or access levels – Prevents abuse by individual users	– Subscription-based models (tiered access) – Personalized service limits for users – User-specific rate control
IP-based rate limiting	Limits the rate of requests from specific IP addresses.	– Focuses on controlling traffic per IP – Ideal for preventing DDoS attacks – Can block or throttle malicious IPs	– Protecting against DDoS attacks – Identifying and blocking malicious traffic – Throttling traffic from untrusted sources

ii. What is an API key? [Manage API keys](#) | [Authentication](#) | [Google Cloud Documentation](#)

e. HTTP source:

https://developer.mozilla.org/en-US/docs/Learn_web_development/Howto/Web_mechanics/What_is_a_web_server

3. Who builds each part of the skeleton, who approves it, and by what date?
 - a. Person A deploys the skeleton
 - i. Person B approves
 - b. Person B creates the endpoints and errors
 - i. Person C approves
 - c. Person C creates documentation for each endpoint detailing what it returns and expects
 - i. Person D approves
 - d. Person D creates automated tests for code to work on every push
 - i. Person A approves
4. Backup
 - a. Rotation

- i. Grayson covers Alayna
- ii. Alayna covers Hunter
- iii. Hunter covers Aidan
- iv. Aidan covers Grayson

5. Teaching the team

- a. Regular meetings?
 - i. Discord call
 - ii. In-person meeting
 - iii. Lunch and learn
- b. Weekly?
 - i. Day undecided